

Residue–Fiber Orthogonality at the Triple–Prime Möbius Gate: Canonical Hilbert Defects, Coherent Synthesis, and Exact Convolution Counterterms

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

TPC–41 separated the terminal physical packet into a closed same–row alias sector and a surviving cross–row four–Möbius sector, while also proving scalar logarithmic decorrelation in reduced residue classes for almost all admissible triple–prime moduli. We determine exactly how far that input can propagate into the physical Hilbert output. For every fixed alias, terminal coefficient vectors belonging to distinct integers in the same residue class are orthogonal. Consequently the natural Hilbert–valued residue energy is identically the already controlled atomic diagonal; the remaining physical term is entirely the coherent synthesis between distinct residue packets. We give a canonical Moore–Penrose factorization into the scalar-visible residue-synchronous sector and a directional defect supported on its domain-orthogonal complement, prove a sharp black-box obstruction for the latter, and reduce each projective mask layer to an orbit–slope Bessel form. A second exact theorem rules out a different shortcut: completing the localized Dirichlet convolution produces boundary faces whose sum is the negative of the original local cell on the physical product window. Thus the shifted-prime completion cancels against the source complement rather than estimating the local packet. These results do not prove the remaining four–Möbius cancellation or breach sieve parity; they replace two broad proposed routes by one explicit cross–residue, orbit–slope coefficient target.

Keywords: Möbius correlation; residue fiber; Hilbert output; Moore–Penrose projection; coherent synthesis; Dirichlet convolution; triple–prime CRT modulus; parity boundary.

Convention. All complex inner products are linear in the first argument.

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1 Introduction

The terminal gate left by TPC–41 has two descriptions that initially appear close. The physical description is a Hilbert norm of row sums whose cross terms contain four Möbius factors. The scalar description is a short-progression variance over reduced residue classes modulo an admissible triple-prime number M , for which TPC–41 proves a logarithmic saving on almost all such moduli [5, 13]. This paper proves that the two descriptions point in transverse directions.

Fix an alias and write the terminal column as

$$\mathcal{T}_k = - \sum_r \mu(r + h + kM) \mathbf{G}_k(r), \quad \mathbf{G}_k(r) \in \ell^2\{(\gamma, j)\}. \quad (1)$$

If $r \neq r'$ but $r \equiv r' \pmod{M}$, then

$$\boxed{\mathbf{G}_k(r) \perp \mathbf{G}_k(r').} \quad (2)$$

Indeed, a shared orbit coordinate would force $(m_\alpha - m_\beta)j$ to be divisible by M ; the admissibility of the auxiliary primes and the no-wrap row window then force the two rows to be identical. Hence the vector energy obtained by packing terms in the same residue class is exactly the atomic diagonal, without using any Möbius cancellation.

This is an orthogonality swap. Arrange the physical atoms in a grid with orbit time j as the horizontal coordinate and target residue modulo M as the vertical coordinate. Along a fixed j , distinct rows occupy distinct residues and the physical output adds them coherently. Along a fixed residue, distinct products occupy disjoint j -coordinates and are orthogonal. Scalar progression variance measures the vertical direction, whereas the surviving physical gate is the horizontal coherent synthesis.

The minimal folded bank admits the exact version

$$\mathcal{E}_{\text{fold,L}} = \sum_s \left\| \sum_a \mathbf{w}_{s,a}^L \right\|^2, \quad \mathcal{E}_{\text{same,L}} = \sum_{s,a} \|\mathbf{w}_{s,a}^L\|^2, \quad (3)$$

where s is a folded alias bucket and a a residue class. Their difference is precisely the cross-residue form

$$\text{Re } \mathcal{F}_L = \text{Re} \sum_s \sum_{a \neq b} \langle \mathbf{w}_{s,a}^L, \mathbf{w}_{s,b}^L \rangle, \quad (4)$$

and similarly on the right. Thus the progression-direction vector variance is already closed by the theorem of TPC-41; what remains is not an unproved Hilbert extension of that scalar statement. The exact packet sums here run over all residues modulo M . The scalar TPC-41 theorem acts only on the unit-residue sector, so nonunit packets remain outside that scalar input unless an independently justified regular pruning makes them vanish.

1.1 The canonical scalar-visible sector

There is nevertheless a precise sense in which any scalar progression theorem can be postcomposed into a Hilbert space. On the sector addressed by that scalar theorem, let R_M be the weighted residue aggregation operator, let A_M synthesize the corresponding physical coefficient vectors, and set

$$P_M = R_M^\dagger R_M, \quad (5)$$

where $R_M^\dagger$ is the Moore–Penrose inverse. Then

$$\boxed{A_M \mu = A_M R_M^\dagger (R_M \mu) + A_M (I - P_M) \mu.} \quad (6)$$

The first term is the maximal canonical sector visible to the scalar progression vector and obeys every bounded Bessel postcomposition of the TPC-41 estimate. The second term is a uniquely defined directional defect. An operator factors through R_M if and only if it vanishes on this defect space. For TPC-41 this canonical split is applied to the unit-residue sector; any nonunit physical packets sit outside that scalar domain and must be carried separately.

This distinction is sharp. On one residue fiber, universal control of a Hilbert synthesis by the scalar fiber sum is possible if and only if all coefficient columns on that fiber are equal. The actual physical columns lie at the opposite geometric extreme: they are pairwise orthogonal. For an unweighted fiber of multiplicity ρ , only the fraction $1/\rho$ of their Hilbert–Schmidt energy lies in the common fiber mode, while $1 - 1/\rho$ lies in the scalar-invisible complement. This rules out a black-box deduction from scalar variance plus coefficient energy. It does not rule out a coefficient-specific theorem for the actual Möbius vector.

TPC–36 adds a useful positive reduction. Each projective mask layer factors its opposite-row coordinate, so its Hilbert norm is exactly an orbit-time Bessel form. The next arithmetic input can therefore be stated for a restricted family of row slopes and orbit weights, rather than for arbitrary Hilbert coefficients.

1.2 The localized convolution counterterm

A second proposed route was to compare the local coefficient $\lambda_L * \mu_D * u_J$ with the completed identity $\Lambda * \mu * \mathbf{1} = \Lambda$. We prove that its natural boundary is not a small error. With

$$A = \lambda_L * \mu_D * u_J, \quad B_D = \lambda_L * (\mu - \mu_D) * \mathbf{1}, \quad B_J = \lambda_L * \mu_D * (\mathbf{1} - u_J), \quad (7)$$

one has the pointwise identity

$$A + B_D + B_J = \lambda_L. \quad (8)$$

The physical product window is of size $L_0 D_0 J \asymp X$, disjoint from the source support $L_0 = o(X)$. Consequently

$$\boxed{B_D + B_J = -A} \quad (9)$$

pointwise on that window, after any Hilbert-valued postcomposition depending only on the grouped product n . The full shifted-prime contribution is cancelled exactly by the source-complement term. For the natural unmasked hard cell we further prove

$$\frac{X}{\log X} \ll \|A\|_2^2 \ll X^{1+o(1)}, \quad A(n)\Lambda(n) = 0, \quad (10)$$

so the local cell and the von Mangoldt comparator have disjoint support. The conclusion is about the completion strategy, not a lower bound for the signed physical four–Möbius output.

1.3 What advances

The present results move the endpoint in three ways. First, they prove that vectorizing the scalar progression direction introduces no hidden arithmetic gain: for the actual packet that direction is exactly diagonal. Second, they replace the vague scalar-to-Hilbert gap by the canonical operator $A_M(I - P_M)$ and then identify its actual cross-residue geometry. Third, they close the local-completion route as a standalone shortcut by an exact counterterm theorem. The remaining positive problem is a coefficient-specific orbit-slope dispersion estimate. No individual affine Chowla estimate, parity breach, Hardy–Littlewood prime-pair asymptotic, or twin-prime conclusion is claimed.

The paper is organized as follows. [Section 2](#) records the inherited endpoint. The next three sections develop the canonical factorization, the residue-fiber identities, and coherent synthesis. The Hilbert large-sieve section records the exact positive scope of dimension-free postcomposition, followed by the projective orbit reduction. The two convolution sections prove the counterterm and its quantitative hard-cell form. The final sections compare the literature, state the next target, and freeze an exact certificate and claim ledger.

2 Inherited terminal interface

We retain only the terminal data proved in TPC–36, TPC–39, and TPC–41 [\[11–13\]](#). Fix $h \in \mathbb{Z} \setminus \{0\}$. A physical source row is

$$\alpha = (\ell_\alpha, d_\alpha), \quad m_\alpha = \ell_\alpha d_\alpha \asymp Q, \quad \ell_\alpha \asymp L_0, \quad d_\alpha \asymp D_0, \quad L_0 D_0 = Q, \quad QJ \asymp X. \quad (11)$$

The source separation makes m_α determine α , and the two inherited source systems obey

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}|^2 \ll X^{o(1)} Q, \quad i = 1, 2. \quad (12)$$

The orbit variable ranges through a set $\mathcal{J}$ of $O(J)$ positive integers.

Choose three pairwise distinct auxiliary primes from the three disjoint TPC-41 intervals, satisfying

$$\max \mathcal{J} < q_i, \quad \max_{\alpha} d_{\alpha} < q_i < \min_{\alpha} \ell_{\alpha}, \quad q_i \nmid h, \quad i = 1, 2, 3, \quad (13)$$

and put $M = q_1 q_2 q_3$. For sufficiently large X , the size and endpoint conditions give

$$(j, M) = 1 \quad (j \in \mathcal{J}), \quad \max_{\alpha, \beta} |m_{\alpha} - m_{\beta}| < M. \quad (14)$$

At the optimized endpoint,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad M = X^{399/400+o(1)}. \quad (15)$$

Let $B \asymp X$ be the completed determinant window. To avoid confusing the source scale L_0 with the alias half-width, write

$$L_B = \left\lfloor \frac{B}{M} \right\rfloor, \quad q = L_B + 1, \quad \mathcal{I}_B = \{-L_B, \dots, L_B\}, \quad K_B = 2L_B + 1. \quad (16)$$

Then

$$K_B = X^{1/400+o(1)} = X^{o(1)} \frac{Q}{J^2}, \quad \frac{Q}{K_B} = X^{o(1)} J^2. \quad (17)$$

For $k \in \mathcal{I}_B$, define

$$n_{\alpha, j, k} = m_{\alpha} j + h + kM, \quad (18)$$

$$b_{\alpha, j, k} = -\mu(n_{\alpha, j, k}) \log(n_{\alpha, j, k}) \Phi_0(n_{\alpha, j, k}), \quad (19)$$

with the second quantity interpreted as zero outside the positive terminal support. The left coefficient and fixed-alias output are

$$c_{\alpha, \gamma, j}^L = \gamma_{\alpha}^{(1)} \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j), \quad (20)$$

$$[\mathcal{T}_k^L]_{\gamma, j} = \sum_{\alpha} c_{\alpha, \gamma, j}^L b_{\alpha, j, k}. \quad (21)$$

The right output interchanges the row systems. The multiplier contains the actual row separation, gcd, residue, and smooth masks, and

$$\sup_{\alpha, \gamma, j} |\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j)| \ll X^{o(1)}. \quad (22)$$

In particular, c^L is independent of k , while its source factor contains $\mu(d_{\alpha})$.

Source separation and the dyadic row window also give the crude counts

$$\#\{\alpha\} + \#\{\gamma\} \ll X^{o(1)} Q. \quad (23)$$

No coprimality of $m_{\alpha} j + h$ with M is assumed in the physical identities below. The scalar theorem of TPC-41 sees only the unit residue sector; keeping all residue classes avoids an unrecorded modulus-dependent pruning of the physical packet.

For the minimal $q = L_B + 1$ Fourier bank, put

$$u_{\alpha, j, 0} = b_{\alpha, j, 0}, \quad (24)$$

$$u_{\alpha, j, a} = b_{\alpha, j, a} + b_{\alpha, j, a-q}, \quad 1 \leq a \leq L_B. \quad (25)$$

The left folded energy and its equal-row part are

$$\mathcal{E}_{\text{fold,L}} = \sum_{\gamma,j} \sum_{a=0}^{L_B} \left| \sum_{\alpha} c_{\alpha,\gamma,j}^L u_{\alpha,j,a} \right|^2, \quad (26)$$

$$\mathcal{E}_{\text{same,L}} = \sum_{\gamma,j} \sum_{a=0}^{L_B} \sum_{\alpha} |c_{\alpha,\gamma,j}^L|^2 |u_{\alpha,j,a}|^2. \quad (27)$$

Writing $\mathcal{F}_L$ for the ordered $\alpha \neq \beta$ cross-row sum,

$$\mathcal{E}_{\text{fold,L}} = \mathcal{E}_{\text{same,L}} + \text{Re } \mathcal{F}_L. \quad (28)$$

TPC-41 proves, together with the symmetric right estimate,

$$\boxed{\mathcal{E}_{\text{same,L}} + \mathcal{E}_{\text{same,R}} \ll X^{o(1)} Q^2 J K_B = X^{o(1)} \frac{Q^3}{J}.} \quad (29)$$

Thus the remaining minimal-bank theorem is exactly the one-sided bound

$$\text{Re}(\mathcal{F}_L + \mathcal{F}_R) \ll X^{o(1)} Q^2 J K_B. \quad (30)$$

3 Canonical scalar factorization and the directional defect

We first isolate the finite-dimensional statement behind any attempted scalar-to-Hilbert upgrade. The result is independent of Möbius cancellation and makes the relevant quantifiers explicit.

Let $\mathcal{A}$ be a finite set of residue labels. For every $a \in \mathcal{A}$, let $\mathcal{I}_a$ be a nonempty finite fiber, and put

$$\mathcal{X} = \bigoplus_{a \in \mathcal{A}} \mathbb{C}^{\mathcal{I}_a}. \quad (31)$$

Choose a nonzero weight vector $w_a = (w_{a,i})_{i \in \mathcal{I}_a}$ on each fiber. Define the weighted scalar collapse

$$(P_w x)_a = \sum_{i \in \mathcal{I}_a} x_{a,i} \overline{w_{a,i}}, \quad P_w : \mathcal{X} \longrightarrow \ell^2(\mathcal{A}). \quad (32)$$

Let $\mathcal{H}$ be a Hilbert space and let $G_{a,i} \in \mathcal{H}$. The corresponding vector synthesis is

$$T_G x = \sum_{a \in \mathcal{A}} \sum_{i \in \mathcal{I}_a} x_{a,i} G_{a,i}. \quad (33)$$

The Moore–Penrose inverse and the orthogonal projection onto $(\ker P_w)^\perp$ are explicit:

$$(P_w^\dagger z)_{a,i} = \frac{z_a w_{a,i}}{\|w_a\|_2^2}, \quad (34)$$

$$\Pi_w := P_w^\dagger P_w, \quad (\Pi_w x)_{a,i} = \frac{w_{a,i}}{\|w_a\|_2^2} \sum_{j \in \mathcal{I}_a} x_{a,j} \overline{w_{a,j}}. \quad (35)$$

Thus every synthesis has the canonical decomposition

$$\boxed{T_G = (T_G P_w^\dagger) P_w + T_G (I - \Pi_w).} \quad (36)$$

The first term is visible to a scalar progression theorem. The second term is the directional defect: it acts on data whose weighted sum is zero in every residue fiber.

Theorem 3.1 (Exact scalar-factorization criterion). *The following statements are equivalent.*

(i) There is a finite constant C such that

$$\|T_G x\|_{\mathcal{H}} \leq C \|P_w x\|_2 \quad (x \in \mathcal{X}). \quad (37)$$

(ii) $\ker P_w \subseteq \ker T_G$.

(iii) The directional defect vanishes:

$$T_G(I - \Pi_w) = 0. \quad (38)$$

(iv) There are vectors $V_a \in \mathcal{H}$ such that

$$G_{a,i} = \overline{w_{a,i}} V_a \quad (a \in \mathcal{A}, i \in \mathcal{I}_a). \quad (39)$$

When these conditions hold,

$$T_G = B P_w, \quad B e_a = V_a, \quad \inf C = \|B\|_{\text{op}} = \|T_G P_w^\dagger\|_{\text{op}}. \quad (40)$$

Proof. Inequality (37) forces $T_G x = 0$ whenever $P_w x = 0$, proving (i) $\Rightarrow$ (ii). Since $I - \Pi_w$ is the orthogonal projection onto $\ker P_w$, conditions (ii) and (iii) are equivalent. On a fixed fiber, (ii) says that the linear map

$$(x_i)_{i \in \mathcal{I}_a} \mapsto \sum_i x_i G_{a,i} \quad (41)$$

vanishes on the kernel of $x \mapsto \sum_i x_i \overline{w_{a,i}}$. A linear map with this kernel property is a vector multiple of that scalar functional, which is exactly (39). Conversely, (39) gives

$$T_G x = \sum_a (P_w x)_a V_a = B P_w x. \quad (42)$$

This proves all equivalences. Because $P_w P_w^\dagger = I$, every factorization $T_G = B P_w$ has $B = T_G P_w^\dagger$; its operator norm is the optimal constant. $\square$

For the unweighted progression collapse, $w_{a,i} = 1$, the criterion says that a universal scalar lift exists precisely when all Hilbert atoms inside the same residue fiber are equal. Pairwise orthogonality, proved for the physical atoms in [section 4](#), lies at the opposite extreme.

Corollary 3.2 (Sharp orthogonal-fiber defect). *Fix one unweighted fiber $\mathcal{I}_a$ of cardinality ρ_a , and suppose that $(G_{a,i})_{i \in \mathcal{I}_a}$ are pairwise orthogonal. Their canonical scalar-synchronous representative is*

$$\overline{G}_a = \frac{1}{\rho_a} \sum_{i \in \mathcal{I}_a} G_{a,i}. \quad (43)$$

Then

$$\sum_{i \in \mathcal{I}_a} \|\overline{G}_a\|^2 = \frac{1}{\rho_a} \sum_{i \in \mathcal{I}_a} \|G_{a,i}\|^2, \quad (44)$$

$$\sum_{i \in \mathcal{I}_a} \|G_{a,i} - \overline{G}_a\|^2 = \left(1 - \frac{1}{\rho_a}\right) \sum_{i \in \mathcal{I}_a} \|G_{a,i}\|^2. \quad (45)$$

Both identities are exact.

Proof. Orthogonality gives

$$\left\| \sum_i G_{a,i} \right\|^2 = \sum_i \|G_{a,i}\|^2. \quad (46)$$

This proves (44). The usual least-squares identity

$$\sum_i \|G_{a,i} - \overline{G}_a\|^2 = \sum_i \|G_{a,i}\|^2 - \rho_a \|\overline{G}_a\|^2 \quad (47)$$

then gives (45). $\square$

Thus, on a fiber of multiplicity ρ_a , only the fraction $1/\rho_a$ of the Hilbert–Schmidt coefficient energy lies in the canonical sector through which a scalar residue sum factors. The remaining fraction is not small when ρ_a grows.

Remark 3.3 (Quantifier boundary). [Theorem 3.1](#) characterizes an implication that is uniform for every scalar array x . Its failure does not say that the one arithmetic array formed by the actual Möbius values has a large directional defect after synthesis. Proving that this particular array avoids the coherent defect would be a coefficient-specific arithmetic theorem. What the criterion rules out is deriving such a theorem solely from a scalar residue norm by a coefficient-blind tensor or Moore–Penrose argument.

4 Exact residue-fiber orthogonality in the physical packet

We now apply the abstract decomposition to the actual terminal vectors. The decisive fact is elementary but rigid: the CRT residue direction and the physical orbit-coordinate direction are transverse.

4.1 Fixed-alias Hilbert atoms

For a retained alias $k \in \mathcal{I}_B$, put

$$A_k(r) = \log(r + h + kM)\Phi_0(r + h + kM), \quad (48)$$

interpreted as zero outside the positive terminal support, and define $\mathbf{G}_k^L(r) \in \ell^2\{(\gamma, j)\}$ by

$$[\mathbf{G}_k^L(r)]_{\gamma, j} = \sum_{\alpha: m_\alpha j = r} c_{\alpha, \gamma, j}^L A_k(r). \quad (49)$$

For fixed (j, r) , at most one row occurs: r/j , when integral, determines m_α , and source separation determines α . Every sum containing $\mu(r + h + kM)$ below is restricted to the positive terminal support, so no value of μ at a nonpositive integer is used. The fixed-alias column is exactly

$$\mathcal{J}_k^L = - \sum_r \mu(r + h + kM) \mathbf{G}_k^L(r). \quad (50)$$

Theorem 4.1 (Physical residue-fiber orthogonality). *Fix $k \in \mathcal{I}_B$. If $r \neq r'$ and*

$$r \equiv r' \pmod{M}, \quad (51)$$

then

$$\langle \mathbf{G}_k^L(r), \mathbf{G}_k^L(r') \rangle = 0. \quad (52)$$

The conclusion is uniform in the terminal cutoff, the Möbius values, and the actual masks. It holds identically for the right vectors.

Proof. A nonzero term in the inner product would require a shared coordinate (γ, j) . There would then be rows α, β such that

$$r = m_\alpha j, \quad r' = m_\beta j. \quad (53)$$

By (51),

$$M \mid (m_\alpha - m_\beta)j. \quad (54)$$

Every $j \in \mathcal{J}$ is positive and smaller than each prime factor of M , so $(j, M) = 1$. Hence $M \mid m_\alpha - m_\beta$. The strict row gap in (14) forces $m_\alpha = m_\beta$, and source separation gives $\alpha = \beta$. Equation (53) then gives $r = r'$, a contradiction. Vanishing cutoffs or coefficients only delete coordinates and cannot invalidate the argument. $\square$

It is convenient to index by the target residue. For every $a \in \mathbb{Z}/M\mathbb{Z}$, define

$$\mathbf{V}_{k,a}^L := - \sum_{r \equiv a-h \pmod{M}} \mu(r+h+kM) \mathbf{G}_k^L(r). \quad (55)$$

Define the actual fixed-alias atomic diagonal

$$\mathcal{D}_{k,L} := \sum_{\gamma,j,\alpha} |c_{\alpha,\gamma,j}^L|^2 |\mu(n_{\alpha,j,k})|^2 |A_k(m_\alpha j)|^2. \quad (56)$$

Corollary 4.2 (Exact packed vector variance). *For every fixed alias,*

$$\mathcal{J}_k^L = \sum_{a \bmod M} \mathbf{V}_{k,a}^L, \quad (57)$$

$$\sum_{a \bmod M} \|\mathbf{V}_{k,a}^L\|^2 = \mathcal{D}_{k,L}, \quad (58)$$

$$\|\mathcal{J}_k^L\|^2 = \mathcal{D}_{k,L} + \sum_{\substack{a,b \bmod M \\ a \neq b}} \langle \mathbf{V}_{k,a}^L, \mathbf{V}_{k,b}^L \rangle. \quad (59)$$

The second sum in (59) is an ordered sum and is real after pairing (a,b) with (b,a) .

Proof. The first identity partitions (50) by target residue. The vectors inside each packet are pairwise orthogonal by theorem 4.1; hence

$$\|\mathbf{V}_{k,a}^L\|^2 = \sum_{r \equiv a-h \pmod{M}} |\mu(r+h+kM)|^2 \|\mathbf{G}_k^L(r)\|^2. \quad (60)$$

When this is summed over a , uniqueness of α for fixed (j,r) turns the right side into (56). Expanding (57) gives the last identity. $\square$

Thus the genuine Hilbert-valued residue variance is already exactly the atomic diagonal. No scalar cancellation is used, and changing any Möbius sign inside one residue fiber leaves (58) unchanged.

The all-residue indexing is essential for the unconditional physical identity. TPC-41's scalar progression theorem is stated on $(\mathbb{Z}/M\mathbb{Z})^\times$, so it controls bounded postcompositions of the unit sector only. If an independently justified regular-packet pruning makes the nonunit packets vanish, the starred version follows as a special case; otherwise the nonunit packets are an additional scalar-invisible sector, not terms that may be silently discarded.

4.2 The complete minimal-folded identity

For a folded bucket $0 \leq s \leq L_B$, define

$$[\mathbf{H}_s^L(r)]_{\gamma,j} := \sum_{\alpha: m_\alpha j = r} c_{\alpha,\gamma,j}^L u_{\alpha,j,s}, \quad (61)$$

and its target-residue packet

$$\mathbf{W}_{s,a}^L := \sum_{r \equiv a-h \pmod{M}} \mathbf{H}_s^L(r). \quad (62)$$

Although $u_{\alpha,j,s}$ contains one or two aliases, both target integers have residue $m_\alpha j + h \pmod{M}$. The proof of theorem 4.1 therefore applies without change to $\mathbf{H}_s^L(r)$.

Theorem 4.3 (Exact folded residue decomposition). *The actual minimal bank satisfies*

$$\mathcal{E}_{\text{same,L}} = \sum_{s=0}^{L_B} \sum_{a \bmod M} \|\mathbf{w}_{s,a}^L\|^2, \quad (63)$$

$$\mathcal{E}_{\text{fold,L}} = \sum_{s=0}^{L_B} \left\| \sum_{a \bmod M} \mathbf{w}_{s,a}^L \right\|^2, \quad (64)$$

$$\mathcal{F}_L = \sum_{s=0}^{L_B} \sum_{\substack{a,b \bmod M \\ a \neq b}} \langle \mathbf{w}_{s,a}^L, \mathbf{w}_{s,b}^L \rangle. \quad (65)$$

The same three identities hold on the right.

Proof. For fixed s , different r 's in one target residue give orthogonal vectors. Therefore

$$\begin{aligned} \sum_a \|\mathbf{w}_{s,a}^L\|^2 &= \sum_r \|\mathbf{H}_s^L(r)\|^2 \\ &= \sum_{\gamma,j,\alpha} |c_{\alpha,\gamma,j}^L|^2 |u_{\alpha,j,s}|^2. \end{aligned} \quad (66)$$

Summing over s proves (63). At a fixed coordinate (γ, j) , summing all target residues is the same as summing all rows, which gives (64). Finally, if $\alpha \neq \beta$ occur at the same (γ, j) , then

$$m_\alpha j + h \not\equiv m_\beta j + h \pmod{M}; \quad (67)$$

otherwise the proof of theorem 4.1 would force $\alpha = \beta$. Consequently the ordered cross-residue terms are exactly the ordered cross-row terms defining $\mathcal{F}_L$, proving (65). $\square$

Combining (63) with the inherited row-diagonal theorem gives the unconditional vector estimate

$$\sum_{s,a} (\|\mathbf{w}_{s,a}^L\|^2 + \|\mathbf{w}_{s,a}^R\|^2) \ll X^{o(1)} Q^2 J K_B. \quad (68)$$

This is a diagonal-scale Hilbert-valued residue theorem for the actual packet. The unresolved operation is no longer packing integers inside one residue; it is the coherent summation over different residues in (64).

5 The coherent residue-synthesis defect

The identities of the preceding section interchange the usual roles of progression and output coordinates. A scalar progression estimate sums down a fixed residue fiber. The physical norm keeps that direction orthogonal and instead sums across distinct residues at each orbit-time coordinate.

5.1 The sharp deterministic overlap bound

For a fixed left alias define the active row overlap

$$R_{k,L} := \max_{\gamma,j} \#\{\alpha : c_{\alpha,\gamma,j}^L b_{\alpha,j,k} \neq 0\}. \quad (69)$$

For the folded bank, let

$$R_{\text{fold,L}} := \max_{\gamma,j,s} \#\{\alpha : c_{\alpha,\gamma,j}^L u_{\alpha,j,s} \neq 0\}. \quad (70)$$

Both are at most the inherited active row count and hence

$$R_{k,L} + R_{\text{fold},L} \ll X^{o(1)}Q. \quad (71)$$

Define the right overlaps symmetrically.

Proposition 5.1 (Deterministic coherent synthesis). *One has*

$$\|\mathcal{T}_k^L\|^2 \leq R_{k,L}\mathcal{D}_{k,L}, \quad (72)$$

$$\mathcal{E}_{\text{fold},L} \leq R_{\text{fold},L}\mathcal{E}_{\text{same},L}. \quad (73)$$

The constants are sharp for abstract packets with the declared overlap numbers.

Proof. Fix (γ, j) . By (67), the nonzero row terms at this coordinate occupy distinct target residues. Cauchy–Schwarz gives

$$\left| \sum_a [\mathbf{V}_{k,a}^L]_{\gamma,j} \right|^2 \leq R_{k,L} \sum_a |[\mathbf{V}_{k,a}^L]_{\gamma,j}|^2. \quad (74)$$

Sum over (γ, j) and use (58). The same argument at every (γ, j, s) , followed by (63)–(64), proves the folded bound. Equality in coordinatewise Cauchy is possible when all active residue contributions have the same phase and modulus, proving abstract sharpness. $\square$

For nonzero packet families (V_a) , write their coherent quotient as

$$\mathfrak{C}(V) := \frac{\|\sum_a V_a\|^2}{\sum_a \|V_a\|^2}. \quad (75)$$

The proposition gives $\mathfrak{C} \leq X^{o(1)}Q$ without arithmetic cancellation. The uniform diagonal-normalized equality-column route has the sufficient benchmark

$$\mathfrak{C}(\mathbf{V}_{0,\bullet}^L \oplus \mathbf{V}_{0,\bullet}^R) \ll X^{o(1)}K_B. \quad (76)$$

Indeed, the inherited fixed-alias diagonal estimate is

$$\mathcal{D}_{0,L} + \mathcal{D}_{0,R} \ll X^{o(1)}Q^2J. \quad (77)$$

The deterministic bound therefore gives only

$$\|\mathcal{T}_0^L\|^2 + \|\mathcal{T}_0^R\|^2 \ll X^{o(1)}Q^3J, \quad (78)$$

whereas the target is

$$K_B Q^2 J = X^{o(1)} \frac{Q^3}{J}. \quad (79)$$

Thus the gap between the deterministic overlap bound and this sufficient endpoint benchmark is

$$\boxed{\frac{Q}{K_B} = X^{o(1)}J^2}. \quad (80)$$

For the complete minimal-folded route, the packed energy is already $O(X^{o(1)}Q^2JK_B)$ by (68). The corresponding uniform diagonal-normalized positive route has the sufficient benchmark of an $X^{o(1)}$ -scale coherent quotient after summing over buckets. The universal overlap estimate loses Q . The logically weakest continuation remains the signed one-sided estimate (30); it need not bound the absolute value of every cross-residue inner product.

5.2 A sharp scalar-blind countermodel

The following model incorporates the exact orthogonality direction of the physical packet. It shows that even perfect scalar cancellation in every residue does not control the coherent synthesis between residues.

Theorem 5.2 (Zero scalar variance with maximal coherent defect). *For every integer $R \geq 1$, there is a two-point atom fiber over each of R residues such that:*

- (i) *the two Hilbert atoms in every residue fiber are orthogonal;*
- (ii) *the scalar sum in every residue is zero;*
- (iii) *every Hilbert coordinate has overlap R ; and*
- (iv) *the coherent synthesis quotient is exactly R .*

More explicitly, let e_+, e_- be an orthonormal basis of $\mathbb{C}^2$, put

$$G_{a,+} = e_+, \quad G_{a,-} = e_-, \quad x_{a,+} = 1, \quad x_{a,-} = -1 \quad (81)$$

for $1 \leq a \leq R$. Then

$$(Px)_a = x_{a,+} + x_{a,-} = 0, \quad (82)$$

$$V_a = x_{a,+}G_{a,+} + x_{a,-}G_{a,-} = e_+ - e_-, \quad (83)$$

$$\sum_a \|V_a\|^2 = 2R, \quad \left\| \sum_a V_a \right\|^2 = 2R^2. \quad (84)$$

The coefficient kernel in this model may be taken constant, so its bounded projective norm is one. Thus low projective complexity does not remove the obstruction. In particular, any universal estimate of the form

$$\left\| \sum_a V_a \right\|^2 \leq C \left(K_B \sum_a \|V_a\|^2 + \|Px\|_2^2 \right) \quad (85)$$

must have

$$C \geq \frac{R}{K_B}. \quad (86)$$

On an abstract dense endpoint packet with $R = X^{o(1)}Q$, this saturates the same J^2 ratio relative to the sufficient benchmark in (80).

There are in fact two independent reasons why the scalar theorem of TPC-41 cannot be inserted as a black box. First, that theorem concerns the complete scalar sum in a residue class; a fixed physical alias keeps only the integers represented by $r = m_\alpha j$, and scalar cancellation is not hereditary under such a deletion. Second, even a hypothetical zero scalar sum on the exact represented subset is compatible with the maximal vector defect in [theorem 5.2](#). The first issue is support; the second is Hilbert geometry.

Remark 5.3 (Arithmetic claim boundary). The countermodel is not asserted to be realized by the actual Möbius values on the TPC packet. It proves that scalar residue variance, residue-fiber orthogonality, bounded projective norm, and the deterministic row ledgers do not logically imply the required physical estimate. A theorem exploiting the actual source signs $\mu(d_\alpha)$, the target signs $\mu(m_\alpha j + h + kM)$, or an averaged auxiliary modulus remains possible and is precisely the coefficient-specific continuation.

6 The tensorizable Hilbert benchmark

There are two logically different ways in which a Hilbert space can enter a large-sieve argument. One may first prove a scalar arithmetic estimate and then apply a bounded synthesis map to its output, or one may insert a vector coefficient before the arithmetic estimate is proved. The first operation is automatic and dimension free. The second is not. This section records the exact distinction at the TPC-41 progression scale.

6.1 Dimension-free postcomposition

Lemma 6.1 (Hilbert tensorization of a scalar operator). *Let I, J be finite sets and let $T = (t_{ji})_{j \in J, i \in I}$ satisfy*

$$\sum_{j \in J} \left| \sum_{i \in I} t_{ji} z_i \right|^2 \leq C^2 \sum_{i \in I} |z_i|^2 \quad (z \in \mathbb{C}^I). \quad (87)$$

Then, for every complex Hilbert space $\mathcal{H}$ and every family $(v_i)_{i \in I} \subset \mathcal{H}$,

$$\sum_{j \in J} \left\| \sum_{i \in I} t_{ji} v_i \right\|_{\mathcal{H}}^2 \leq C^2 \sum_{i \in I} \|v_i\|_{\mathcal{H}}^2. \quad (88)$$

The constant is independent of $\dim \mathcal{H}$, including when $\mathcal{H}$ is infinite dimensional.

Proof. All the vectors involved lie in a separable closed subspace. Expand them against an orthonormal basis (e_ν) , apply (87) to the scalar coefficients $\langle v_i, e_\nu \rangle$, and sum in ν . Parseval and Tonelli give (88) with the same constant. $\square$

For example, tensorizing the classical multiplicative large sieve [10] gives, for finitely supported $v_n \in \mathcal{H}$,

$$\sum_{r \leq R} \frac{r}{\varphi(r)} \sum_{\chi \bmod r}^* \left\| \sum_{n \leq Z} v_n \chi(n) \right\|_{\mathcal{H}}^2 \ll (Z + R^2) \sum_{n \leq Z} \|v_n\|_{\mathcal{H}}^2. \quad (89)$$

The usual dyadic decomposition of initial intervals, now using Cauchy–Schwarz in $\mathcal{H}$, also gives the maximal benchmark

$$\sum_{r \leq R} \frac{r}{\varphi(r)} \sum_{\chi \bmod r}^* \max_{z \leq Z} \left\| \sum_{n \leq z} v_n \chi(n) \right\|_{\mathcal{H}}^2 \ll (Z + R^2) (\log(2Z))^2 \sum_{n \leq Z} \|v_n\|_{\mathcal{H}}^2. \quad (90)$$

These are standard tensor consequences of scalar large-sieve estimates; they are recorded as a benchmark, not claimed as a new vector-valued large sieve.

6.2 Simultaneous Bessel transfer of the TPC-41 vector

Let $\mathfrak{M}_X$ be the TPC-41 family obtained by taking one prime from each of three fixed, pairwise disjoint, fixed-proportion intervals at scale $X^{133/400}$. For $M \in \mathfrak{M}_X$, write

$$\mathcal{U}_M = (\mathbb{Z}/M\mathbb{Z})^\times, \quad K_M = X/M, \quad (91)$$

and, for a family of bounded-variation weights $W = W_X$, supported in a fixed compact subinterval of $(0, \infty)$ and with uniformly bounded BV norm, define

$$S_{M,W}(a) = \sum_{n \equiv a \pmod{M}} \mu(n) W(n/X), \quad a \in \mathcal{U}_M. \quad (92)$$

TPC-41 proves that, for every fixed $0 < \theta < 1/200$ and every fixed finite collection of such uniformly bounded weights, all but $o(|\mathfrak{M}_X|)$ moduli satisfy simultaneously

$$\|S_{M,W}\|_{\ell^2(\mathcal{U}_M)}^2 \ll_{\theta,W} \frac{XK_M}{(\log X)^\theta}. \quad (93)$$

See Wang [13]; the underlying centered progression input is due to Klurman et al. [5].

Proposition 6.2 (Simultaneous Bessel postcomposition). *Fix a modulus M for which (93) holds. Let $\mathcal{H}_M$ be any complex Hilbert space and let*

$$B_M : \ell^2(\mathcal{U}_M) \longrightarrow \mathcal{H}_M \quad (94)$$

be any bounded linear map. Then

$$\boxed{\|B_M S_{M,W}\|_{\mathcal{H}_M}^2 \ll_{\theta,W} \|B_M\|_{\text{op}}^2 \frac{XK_M}{(\log X)^\theta}.} \quad (95)$$

The same good modulus works simultaneously for every choice of $\mathcal{H}_M$ and B_M ; in particular, B_M may be selected after the scalar good set has been fixed.

Equivalently, suppose that $v_{M,a} \in \mathcal{H}_M$ has Bessel synthesis bound

$$\left\| \sum_{a \in \mathcal{U}_M} z_a v_{M,a} \right\|_{\mathcal{H}_M}^2 \leq \beta_M \sum_{a \in \mathcal{U}_M} |z_a|^2. \quad (96)$$

Then

$$\left\| \sum_{\substack{n \geq 1 \\ (n,M)=1}} \mu(n) W(n/X) v_{M,n \bmod M} \right\|_{\mathcal{H}_M}^2 \ll_{\theta,W} \beta_M \frac{XK_M}{(\log X)^\theta}. \quad (97)$$

Proof. The first assertion is simply $\|B_M S_{M,W}\| \leq \|B_M\|_{\text{op}} \|S_{M,W}\|_2$, followed by (93). No exceptional set is introduced by this deterministic inequality. For the second assertion, take $B_M e_a = v_{M,a}$. Then (96) says $\|B_M\|_{\text{op}}^2 \leq \beta_M$, and expanding $B_M S_{M,W}$ gives (97). $\square$

For the terminal logarithmic atom, take a fixed smooth compactly supported Φ and

$$W_X(t) = \frac{\log(Xt)}{\log X} \Phi(t).$$

This family has uniformly bounded variation. Multiplying (95) by $(\log X)^2$ yields

$$\left\| B_M \left(\sum_{\substack{n \equiv a \\ (\bmod M)}} \mu(n) \log n \Phi(n/X) \right) \right\|_{\mathcal{H}_M}^2 \ll_{\theta,\Phi} \|B_M\|_{\text{op}}^2 \frac{XK_M (\log X)^2}{(\log X)^\theta}. \quad (98)$$

The qualifier *simultaneous* is useful here. Even if the physical projective decomposition contains many residue masks or Hilbert output layers, no union bound over those layers is needed when they are all bounded postcompositions of the same vector $S_{M,W}$. Their entire cost is the appropriate operator or Bessel norm.

6.3 The precise stopping point

[Proposition 6.2](#) applies when the coefficient attached to n is residue synchronous:

$$G_M(n) = W(n/X)v_{M,n \bmod M}. \quad (99)$$

It does not estimate

$$\sum_n \mu(n)G_M(n) \quad (100)$$

for an arbitrary field $G_M(n) \in \mathcal{H}_M$ that changes direction among integers in the same residue fiber. Inserting $G_M(n)$ before applying the scalar theorem changes its coefficient sequence, whereas postcomposition changes only its already formed residue output.

This also explains why the terminology *vector-valued KMT theorem* would be inaccurate. The logarithmic saving in [\(93\)](#) first uses the scalar multiplicativity and pretentious structure in KMT to identify and subtract one most-correlated character; TPC-41 then removes that rank-one term on average over the declared modulus family. A general map $n \mapsto G_M(n)$ has no such multiplicative structure. Only the postcomposition step is a dimension-free tensorization. The within-fiber directional component left after the canonical residue projection therefore requires a genuinely different arithmetic input at the desired saving; it is not an omitted application of [\(89\)](#); compare the exact factorization criterion in [theorem 3.1](#).

7 Projective reduction to an orbit-slope Bessel theorem

The residue obstruction does not make the low projective complexity of the physical masks useless. It tells us exactly what that complexity can and cannot reduce. We now pass from the full Hilbert coefficient to one declared arithmetic form on each TPC-36 layer.

7.1 Exact one-layer norm identity

Fix one of the fixed number of orbit residue cells produced in TPC-36 and replace $\mathcal{J}$ by its intersection with that cell; we keep the notation $\mathcal{J}$. On this fixed cell TPC-36 gives the exact representation

$$\mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) = \int_{\Omega} p_{\omega}(\alpha)q_{\omega}(\gamma)\eta_{\omega}(j/J) d\nu(\omega), \quad (101)$$

and, for every fixed smoothness order needed below,

$$\int_{\Omega} \|p_{\omega}\|_{\infty}\|q_{\omega}\|_{\infty}\|\eta_{\omega}\|_{C^B} d|\nu|(\omega) \ll_B X^{o(1)}. \quad (102)$$

The factors may be normalized to have sup norm at most one, with their sizes absorbed into $d|\nu|$. Put

$$\Gamma_{\omega,\alpha} := \gamma_{\alpha}^{(1)}p_{\omega}(\alpha). \quad (103)$$

The left output of one layer at alias k is

$$[\mathcal{T}_{\omega,k}^L]_{\gamma,j} = q_{\omega}(\gamma)\eta_{\omega}(j/J) \sum_{\alpha} \Gamma_{\omega,\alpha} b_{\alpha,j,k}. \quad (104)$$

Proposition 7.1 (Exact projective layer reduction). *For every projective layer,*

$$\|\mathcal{T}_{\omega,k}^L\|_2^2 = \|q_{\omega}\|_2^2 \sum_{j \in \mathcal{J}} |\eta_{\omega}(j/J)|^2 \left| \sum_{\alpha} \Gamma_{\omega,\alpha} b_{\alpha,j,k} \right|^2, \quad (105)$$

$$\sum_{k \in \mathcal{I}_B} \|\mathcal{T}_{\omega,k}^L\|_2^2 = \|q_{\omega}\|_2^2 \sum_{j \in \mathcal{J}} |\eta_{\omega}(j/J)|^2 \sum_{k \in \mathcal{I}_B} \left| \sum_{\alpha} \Gamma_{\omega,\alpha} b_{\alpha,j,k} \right|^2. \quad (106)$$

The corresponding layer diagonals are

$$\mathcal{D}_{\omega,k}^L = \|q_\omega\|_2^2 \sum_j |\eta_\omega(j/J)|^2 \sum_\alpha |\Gamma_{\omega,\alpha}|^2 |b_{\alpha,j,k}|^2, \quad (107)$$

$$\mathcal{D}_{\omega,\text{tr}}^L = \sum_{k \in \mathcal{I}_B} \mathcal{D}_{\omega,k}^L. \quad (108)$$

Proof. In (104), the dependence on γ is the single factor $q_\omega(\gamma)$. Square, sum first over γ , and then over j . This proves (105); summing over k proves the trace identity. Deleting the unequal-row products inside the same squares gives (107) and (108). $\square$

The proposition shows that Hilbert dualization introduces no mysterious new coordinate after projective separation. It leaves the explicit orbit-slope form

$$\sum_{j \in \mathcal{J}} w_\omega(j) \left| \sum_\alpha \Gamma_{\omega,\alpha} \mu(m_\alpha j + h + kM) A_k(m_\alpha j) \right|^2, \quad w_\omega(j) = |\eta_\omega(j/J)|^2. \quad (109)$$

For fixed j , all rows in this sum occupy distinct target residues modulo M . It is therefore a transversal row estimate, not a variance inside one arithmetic progression.

The same conclusion follows directly by duality. If $z = (z_{\gamma,j})$ is a unit vector, then a layer contributes

$$\sum_j \eta_\omega(j/J) \left(\sum_\gamma q_\omega(\gamma) \overline{z_{\gamma,j}} \right) \sum_\alpha \Gamma_{\omega,\alpha} b_{\alpha,j,k}, \quad (110)$$

and

$$\sum_j \left| \sum_\gamma q_\omega(\gamma) \overline{z_{\gamma,j}} \right|^2 \leq \|q_\omega\|_2^2 \|z\|_2^2. \quad (111)$$

Conversely, every orbit vector in the direction of q_ω is attained. Thus (105) is the exact duality norm, not a Cauchy majorant.

7.2 The coefficient-specific continuation

The direct equality route would follow from the following layer-stable estimate:

$$\begin{aligned} \sum_j |\eta_\omega(j/J)|^2 \left| \sum_\alpha \Gamma_{\omega,\alpha} b_{\alpha,j,0} \right|^2 \\ \ll X^{o(1)} K_B \sum_j |\eta_\omega(j/J)|^2 \sum_\alpha |\Gamma_{\omega,\alpha}|^2 |b_{\alpha,j,0}|^2. \end{aligned} \quad (112)$$

The cleaner full-trace route asks for

$$\begin{aligned} \sum_{j,k} |\eta_\omega(j/J)|^2 \left| \sum_\alpha \Gamma_{\omega,\alpha} b_{\alpha,j,k} \right|^2 \\ \ll X^{o(1)} \sum_{j,k} |\eta_\omega(j/J)|^2 \sum_\alpha |\Gamma_{\omega,\alpha}|^2 |b_{\alpha,j,k}|^2. \end{aligned} \quad (113)$$

Here and below the quantifier ranges only over the projective layers actually generated by TPC-36, their fixed residue cells, and the two physical row systems. It does not range over arbitrary bounded source coefficients.

Proposition 7.2 (Uniform layer theorem implies the endpoint bounds). *If (112) holds uniformly on all actual normalized layers and residue cells, then*

$$\|\mathcal{T}_0^L\|^2 + \|\mathcal{T}_0^R\|^2 \ll X^{o(1)} K_B Q^2 J. \quad (114)$$

If (113) holds uniformly instead, then

$$\sum_{k \in \mathcal{I}_B} (\|\mathcal{T}_k^L\|^2 + \|\mathcal{T}_k^R\|^2) \ll X^{o(1)} Q^2 J K_B. \quad (115)$$

These are absolute endpoint bounds; no comparison with the atomic diagonal of the signed reassembled kernel is asserted.

Proof. Normalize $p_\omega, q_\omega, \eta_\omega$ to have sup norm at most one and absorb their former sizes into the signed layer measure. Then its total variation is $P = X^{o(1)}$ by (102). Moreover,

$$\|q_\omega\|_2^2 \ll X^{o(1)} Q, \quad \sum_{\alpha} |\Gamma_{\omega, \alpha}|^2 \ll X^{o(1)} Q, \quad |\mathcal{J}| \ll J, \quad (116)$$

by (12), (23), and the normalization. The terminal logarithms cost only $X^{o(1)}$.

The equality-layer hypothesis and [proposition 7.1](#) therefore give

$$\|\mathcal{T}_{\omega, 0}^L\|^2 \ll X^{o(1)} K_B Q^2 J \quad (117)$$

uniformly. Minkowski's integral inequality costs $P^2 = X^{o(1)}$ after squaring and proves the left part of (114). Interchanging the row systems gives the right part.

For the trace hypothesis, work in $\ell^2(\mathcal{I}_B) \otimes \mathcal{H}$. The same three ledgers, now with $|\mathcal{I}_B| = K_B$, give

$$\sum_k \|\mathcal{T}_{\omega, k}^L\|^2 \ll X^{o(1)} Q^2 J K_B. \quad (118)$$

Minkowski in this product Hilbert space again costs only P^2 , with no additional alias factor. The fixed residue-cell reassembly is absorbed in $X^{o(1)}$, completing the proof. $\square$

Normalization before applying the uniform layer theorem is necessary. A bound proved separately after renormalizing each formal layer, followed by an absolute count of the $O(D_0)$ Ferrers layers, would reintroduce a polynomial loss. TPC-36 supplies small signed projective mass, not a small raw layer count.

Remark 7.3 (What projective reduction does not prove). The constant kernel used in [theorem 5.2](#) is itself a projective layer of norm one. Hence (102) alone implies neither (112) nor (113). The new input must exploit the actual arithmetic direction $\Gamma_{\omega, \alpha}$, including its source Möbius sign, jointly with the target Möbius values along $m_\alpha j + h + kM$. Establishing that restricted Bessel theorem would be genuine progress beyond scalar progression variance; its statement does not claim a parity breach or a prime-pair asymptotic.

8 Localized convolution completion is an exact counterterm

The full identity $\Lambda * \mu * \mathbf{1} = \Lambda$ does not make one dyadic factorization cell an approximation to Λ . In the present scales the precise opposite statement holds: the complementary divisor faces sum to the negative of the local cell on the physical product window. We formulate this before inserting any arithmetic estimate.

8.1 The three-face identity

Let δ_1 denote the identity for Dirichlet convolution. Let λ_L be any finitely supported arithmetic function, let μ_D be any finitely supported truncation of μ , and let u_J be any finitely supported orbit cutoff. In the physical specialization,

$$\lambda_L(p) = (\log p)w_L(p/L_0) \quad (p \text{ prime}), \quad \mu_D(d) = \mu(d)w_D(d/D_0), \quad u_J(j) = w_J(j/J), \quad (119)$$

and λ_L vanishes away from the prime source interval. Define

$$A := \lambda_L * \mu_D * u_J, \quad (120)$$

$$B_D := \lambda_L * (\mu - \mu_D) * \mathbf{1}, \quad (121)$$

$$B_J := \lambda_L * \mu_D * (\mathbf{1} - u_J). \quad (122)$$

The complements in (121) and (122) need not be finitely supported. Every convolution value at a fixed integer is nevertheless a finite divisor sum.

Theorem 8.1 (Exact three-face completion). *For every positive integer n ,*

$$\boxed{A(n) + B_D(n) + B_J(n) = \lambda_L(n)}. \quad (123)$$

Proof. Using $\mu * \mathbf{1} = \delta_1$ and associativity gives

$$\begin{aligned} A + B_D + B_J &= \lambda_L * [\mu_D * u_J + (\mu - \mu_D) * \mathbf{1} + \mu_D * (\mathbf{1} - u_J)] \\ &= \lambda_L * (\mu * \mathbf{1}) = \lambda_L * \delta_1 = \lambda_L. \end{aligned}$$

□

The ordering of the two boundary terms is deliberate. For hard cutoffs, B_D first takes the complementary d -range with all j , while B_J retains the physical d -range and takes the complementary j -range. Thus there is no overlap ambiguity. Smooth cutoffs give the same identity with a weighted partition.

8.2 The physical-window counterterm

Assume that, for fixed positive dyadic constants,

$$\text{supp } \lambda_L \subset [c_L L_0, C_L L_0], \quad \text{supp } \mu_D \subset [c_D D_0, C_D D_0], \quad \text{supp } u_J \subset [c_J J, C_J J], \quad (124)$$

where

$$L_0 D_0 J \asymp X, \quad D_0 J \longrightarrow \infty. \quad (125)$$

There are fixed $0 < c_X < C_X$ such that the local convolution A is supported in

$$W_X = [c_X X, C_X X] \cap \mathbb{N}. \quad (126)$$

For all sufficiently large X , this window is disjoint from $\text{supp } \lambda_L$.

Corollary 8.2 (The boundary is the negative local cell). *On the physical product window,*

$$\boxed{\mathbf{1}_{W_X}(B_D + B_J) = -\mathbf{1}_{W_X}A}. \quad (127)$$

Consequently,

$$\|B_D + B_J\|_{\ell^2(W_X)} = \|A\|_{\ell^2(W_X)}, \quad (128)$$

$$\|B_D\|_{\ell^2(W_X)}^2 + \|B_J\|_{\ell^2(W_X)}^2 \geq \frac{1}{2} \|A\|_{\ell^2(W_X)}^2. \quad (129)$$

In particular, at least one of the two boundary faces has squared norm at least $\|A\|_{\ell^2(W_X)}^2/4$.

Proof. Restrict (123) to W_X , where λ_L vanishes. This proves (127) and (128). The inequality $\|x + y\|^2 \leq 2\|x\|^2 + 2\|y\|^2$ proves (129). $\square$

The counterterm statement is unchanged by terminal multiplication and Hilbert postcomposition.

Proposition 8.3 (Arbitrary Hilbert multiplier). *This proposition concerns the grouped-product scalarization; it does not identify a representation-dependent physical mask with a function of the grouped product n . Let $\mathcal{H}$ be a Hilbert space and let $v : W_X \rightarrow \mathcal{H}$ be arbitrary. All sums below are finite. Then*

$$\boxed{\sum_{n \in W_X} (B_D + B_J)(n)v(n) = - \sum_{n \in W_X} A(n)v(n).} \quad (130)$$

This remains true for each admissible alias after taking, on the positive terminal support,

$$v(n) = \mu(n + h + kM) \log(n + h + kM) \Phi_0(n + h + kM) z(n), \quad z(n) \in \mathcal{H}. \quad (131)$$

Proof. Multiply the pointwise identity (127) by $v(n)$ and sum. No boundedness or regularity of v is required because W_X is finite. $\square$

Thus an estimate for the combined boundary output in (130) is exactly an estimate for the original localized output. Completion has not replaced the local gate by a smaller error.

8.3 Why the completed Λ term cancels

The same calculation locates the apparent shifted-prime term. Since $f * \mu * \mathbf{1} = f$ for every arithmetic function f ,

$$\boxed{\Lambda = (\Lambda - \lambda_L) + A + B_D + B_J.} \quad (132)$$

Indeed, the first term is $(\Lambda - \lambda_L) * \mu * \mathbf{1}$, while the other three terms are the three-face expansion of $\lambda_L * \mu * \mathbf{1}$.

Corollary 8.4 (Source-complement cancellation). *On the physical window one has the two separate identities*

$$\mathbf{1}_{W_X}(\Lambda - \lambda_L) = \mathbf{1}_{W_X} \Lambda, \quad (133)$$

$$\mathbf{1}_{W_X}(A + B_D + B_J) = 0. \quad (134)$$

Thus, after multiplication by any grouped-product Hilbert multiplier and summation, the completed Λ contribution is carried entirely by the source-complement term, while the three local faces cancel among themselves. Equivalently, moving the local cell to the other side leaves $A = -(B_D + B_J)$.

This is an algebraic obstruction, not a statement that shifted-prime estimates are weak. Even a perfect bound for the completed Λ term or its source-complement copy gives no estimate for one local face: the second identity in (134) only cancels the three faces collectively.

8.4 Compatibility with projective mask layers

The actual row mask is representation-dependent, so it cannot simply be inserted as a function of the grouped product n in (123). After the product-ready fixed-column projective separation inherited from TPC-36, however, each scalarized layer has the form

$$A_\rho = B_\rho * F_\rho * E_\rho, \quad (135)$$

where the three functions carry the source, divisor, and orbit weights of that layer. There is then a layerwise version of the same obstruction.

Proposition 8.5 (Unique layerwise completion). *Put*

$$C_\rho = F_\rho * E_\rho, \quad R_\rho = \delta_1 - C_\rho. \quad (136)$$

Then R_ρ is the unique arithmetic function for which $C_\rho + R_\rho = \delta_1$, and

$$A_\rho + B_\rho * R_\rho = B_\rho. \quad (137)$$

Consequently, on any window W disjoint from $\text{supp } B_\rho$,

$$\boxed{\mathbf{1}_W(B_\rho * R_\rho) = -\mathbf{1}_W A_\rho.} \quad (138)$$

Proof. The uniqueness of R_ρ is immediate from (136). Associativity gives

$$A_\rho + B_\rho * R_\rho = B_\rho * (C_\rho + R_\rho) = B_\rho * \delta_1 = B_\rho.$$

Restriction to W proves (138). $\square$

A low projective reassembly mass controls the cost of summing the layers; it does not make the exact counterterms in (138) small. Moreover, the scalar convolution in (135) has already summed the orbit variable. The physical output retains its (γ, j) coordinates, so the scalar completion identity does not by itself define an inverse on that Hilbert coordinate space. The results of this section therefore rule out a small-boundary shortcut; they do not supply, or disprove, the remaining physical cancellation theorem.

9 A quantitative lower bound for the localization boundary

The preceding counterterm identity is independent of coefficient size. For the natural unmasked hard cell one can also prove that the local coefficient, and hence at least one boundary face, has genuinely macroscopic ℓ^2 -energy. We state every support hypothesis because the lower bound is not asserted for arbitrary physical masks, which may vanish on a chosen subfamily.

9.1 Separated hard intervals

Let

$$I_L = [a_L L_0, b_L L_0] \cap \mathbb{N}, \quad I_D = [a_D D_0, b_D D_0] \cap \mathbb{N}, \quad I_J = [a_J J, b_J J] \cap \mathbb{N}, \quad (139)$$

where all endpoints are fixed positive constants with $a_S < b_S$. Assume that, for all sufficiently large X ,

$$2 \leq \min I_D, \quad \max I_D < \min I_J, \quad \max I_J < \min I_L, \quad (140)$$

and

$$L_0 D_0 J \asymp X, \quad \log L_0 \asymp \log D_0 \asymp \log J \asymp \log X. \quad (141)$$

These conditions hold at the inherited endpoint: TPC-36 gives $D_0 = X^{10049/52500+o(1)}$ and $L_0 = X^{99979/210000+o(1)}$, hence $D_0 = o(J)$ and $J = o(L_0)$ [11]. Define the hard arithmetic functions

$$\lambda_L(n) = (\log n) \mathbf{1}_{n \in I_L \cap \mathbb{P}}, \quad \mu_D(n) = \mu(n) \mathbf{1}_{n \in I_D}, \quad u_J(n) = \mathbf{1}_{n \in I_J}, \quad (142)$$

and put

$$C = \mu_D * u_J, \quad A = \lambda_L * C. \quad (143)$$

Lemma 9.1 (Outer-prime uniqueness and support separation). *If*

$$\ell_1 e_1 = \ell_2 e_2, \quad \ell_i \in I_L \cap \mathbb{P}, \quad e_i = d_i j_i, \quad d_i \in I_D, \quad j_i \in I_J, \quad (144)$$

then $\ell_1 = \ell_2$ and $e_1 = e_2$. Moreover,

$$\boxed{A(n)\Lambda(n) = 0 \quad (n \geq 1).} \quad (145)$$

Proof. If $\ell_1 \neq \ell_2$, then the primality of the two source variables and (144) imply $\ell_1 \mid d_2 j_2$. Hence $\ell_1 \mid d_2$ or $\ell_1 \mid j_2$, which is impossible because $d_2, j_2 < \ell_1$ by (140). This proves uniqueness.

If a summand contributing to $A(n)$ is nonzero, then $n = \ell d j$, where ℓ is prime and $d \geq 2$ with $\mu(d) \neq 0$. Any prime $p \mid d$ satisfies $p \leq d < \ell$, so $p \neq \ell$. Thus n has at least two distinct prime divisors and is not a prime power. Therefore $\Lambda(n) = 0$, proving (145). $\square$

Outer-prime uniqueness gives an exact tensor factorization rather than only an upper bound.

Proposition 9.2 (Exact coefficient-energy factorization). *Under (140),*

$$\boxed{\|A\|_2^2 = \left(\sum_{\ell \in I_L \cap \mathbb{P}} (\log \ell)^2 \right) \left(\sum_e |C(e)|^2 \right).} \quad (146)$$

Proof. For every e in the support of C and every source prime ℓ , one has $A(\ell e) = (\log \ell)C(e)$. By lemma 9.1, distinct pairs (ℓ, e) give distinct products. Squaring and summing proves the identity. $\square$

9.2 Endpoint-size lower and upper bounds

Theorem 9.3 (Energy of the natural localized cell). *Under (139)–(141),*

$$\|A\|_2^2 \gg L_0 D_0 J \frac{\log L_0}{\log D_0 \log J} \gg \frac{X}{\log X}, \quad (147)$$

while

$$\|A\|_2^2 \ll X^{1+o(1)}. \quad (148)$$

The implied constants in the lower bound depend only on the fixed interval endpoints.

Proof. Choose fixed positive-width interior intervals $I_D^\circ \subset I_D$ and $I_J^\circ \subset I_J$. If $p_D \in I_D^\circ$ and $p_J \in I_J^\circ$ are prime, then the only divisor of $p_D p_J$ lying in I_D whose complementary divisor lies in I_J is p_D . Indeed, the four divisors are $1, p_D, p_J, p_D p_J$, and the scale separation excludes the other three. Consequently,

$$C(p_D p_J) = \mu(p_D) = -1. \quad (149)$$

Because the two prime intervals are strictly separated, the smaller prime factor of $p_D p_J$ is uniquely p_D ; distinct prime pairs therefore give distinct integers e . The prime number theorem therefore gives

$$\sum_e |C(e)|^2 \gg \frac{D_0 J}{\log D_0 \log J}. \quad (150)$$

It also gives

$$\sum_{\ell \in I_L \cap \mathbb{P}} (\log \ell)^2 \asymp L_0 \log L_0. \quad (151)$$

Inserting (150) and (151) into (146) proves (147).

For the upper bound, let

$$r(e) = \#\{d \in I_D : d \mid e, e/d \in I_J\}. \quad (152)$$

Cauchy–Schwarz gives

$$|C(e)|^2 \leq r(e) \sum_{\substack{d \in I_D \\ d|e \\ e/d \in I_J}} \mu(d)^2. \quad (153)$$

Now $r(e) \leq \tau(e) = X^{o(1)}$ uniformly for $e \ll D_0 J \ll X^{O(1)}$. Summing (153) over e yields

$$\sum_e |C(e)|^2 \ll X^{o(1)} D_0 J. \quad (154)$$

Together with (151) and $L_0 D_0 J \asymp X$, this proves (148). $\square$

Let B_D, B_J be the boundary faces from (121)–(122), formed from the hard functions in (142), and let W_X contain the product support of A while remaining disjoint from I_L .

Corollary 9.4 (A boundary face has macroscopic energy). *One has*

$$\|B_D\|_{\ell^2(W_X)}^2 + \|B_J\|_{\ell^2(W_X)}^2 \geq \frac{1}{2} \|A\|_2^2 \gg \frac{X}{\log X}, \quad (155)$$

$$\max\{\|B_D\|_{\ell^2(W_X)}^2, \|B_J\|_{\ell^2(W_X)}^2\} \geq \frac{1}{4} \|A\|_2^2 \gg \frac{X}{\log X}. \quad (156)$$

Proof. Combine corollary 8.2 with theorem 9.3. $\square$

There is also an exact orthogonality obstruction to approximating the cell by a localized von Mangoldt function. Let $W_X^\sharp = [aX, bX] \cap \mathbb{N}$, with fixed $0 < a < b$, contain the support of A .

Corollary 9.5 (Orthogonal mismatch with von Mangoldt). *One has*

$$\|\mathbf{1}_{W_X^\sharp} (A - \Lambda)\|_2^2 = \|A\|_2^2 + \|\mathbf{1}_{W_X^\sharp} \Lambda\|_2^2. \quad (157)$$

In particular,

$$\|\mathbf{1}_{W_X^\sharp} (A - \Lambda)\|_2^2 \gg X \log X. \quad (158)$$

Proof. Equation (145) makes the two coefficient sequences pointwise orthogonal, proving (157). The prime number theorem in the form

$$\sum_{aX \leq n \leq bX} \Lambda(n)^2 \asymp X \log X$$

proves (158). $\square$

9.3 Claim boundary

The lower bounds in this section concern the natural unmasked hard cell. They are not asserted for every projective layer or for the fully masked physical packet: a unary weight or residue factor may vanish on the prime subfamily used in (149). Likewise, coefficient ℓ^2 -energy is not a lower bound for the signed terminal correlation after multiplication by $\mu(n + h + kM)$, and its scalar normalization is not the normalization of the full (γ, j) -Hilbert output.

Accordingly, theorem 9.3 does not show that the physical gate is false or that its Möbius factors fail to cancel. It proves the narrower and methodologically decisive fact that the localization boundary is not a deterministically small approximation error. Any successful continuation must estimate the localized coefficient system itself, or its canonical residue-fiber defect, rather than importing a completed shifted-prime theorem through (132). No fixed-shift Chowla, prime-pair asymptotic, parity breach, or twin-prime conclusion is claimed.

10 Sparse families, dense shifts, and the exact selection loss

We place the preceding operator statement against three neighboring parts of the literature. The purpose is not to claim a new general large sieve, but to identify which existing average can actually select an admissible TPC modulus or shift.

10.1 Sparse moduli: what is already available

Large-sieve and Bombieri–Vinogradov theory for sparse sets of moduli is well established. Baier’s general sparse large sieve requires quantitative distribution of the modulus set in arithmetic progressions [1]. Baier and Zhao prove, among other results, a Barban–Davenport–Halberstam theorem for a well-distributed set $\mathcal{S}$ satisfying a divisor-slice stability condition and

$$|\mathcal{S}(Q)| \gg Q^{1/2}, \quad Q \leq x^{1-\varepsilon}; \quad (159)$$

see Baier and Zhao [2, Theorem 1]. These hypotheses are substantive, not consequences of cardinality alone.

For the three disjoint prime intervals used here,

$$|\mathfrak{M}_X| \asymp \frac{M_0}{(\log X)^3}, \quad M_0 = X^{399/400}, \quad (160)$$

so the numerical size condition in (159) is more than satisfied, and $M_0 \leq X^{1-\varepsilon}$ for every fixed $0 < \varepsilon < 1/400$. A direct invocation would still require a proof that this exact three-interval product family is well distributed and satisfies the divisor-slice hypothesis. There is also a quantifier issue: the cited formulation starts from one fixed set of natural numbers, whereas the three prime intervals here move with X . One must either embed the moving slices in a fixed set or verify that the proof is uniform for this scale-dependent family; a finite dyadic decomposition is also needed if the product window crosses dyadic boundaries. Moreover, the cited Barban–Davenport–Halberstam theorem concerns the scalar prime error. The standard sparse large sieve likewise acts on one common scalar coefficient sequence. Neither statement permits the modulus-dependent within-fiber physical coefficient field encoded by T_G and its canonical projector to be inserted into that scalar sequence for free.

Two further comparisons prevent an inflated novelty claim. Iwaniec obtains true-order large-sieve estimates for prime moduli when the common scalar coefficients are supported on integers without small prime factors [4]; this is not a theorem for products of three primes with moving Hilbert coefficients. As a recent comparison for factorable moduli, Li [6] proves prime mean-value theorems for bilinear modulus forms up to $x^{9/17}$ and trilinear modulus forms up to $x^{17/32}$. Those results concern scalar prime distribution and their trilinear exponent is far below $399/400$. Accordingly, we claim neither a first sparse triple-prime BDH theorem nor a new general almost-prime-modulus large sieve.

10.2 An exact density-loss lemma

The elementary loss incurred by restricting a dense average is already decisive at the present logarithmic density.

Lemma 10.1 (Sharp restriction threshold). *Let Ω_X be a finite ambient family, let $\mathcal{S}_X \subseteq \Omega_X$ have relative density $\delta_X = |\mathcal{S}_X|/|\Omega_X|$, and let $E_X : \Omega_X \rightarrow [0, \infty)$. Then*

$$\frac{1}{|\mathcal{S}_X|} \sum_{s \in \mathcal{S}_X} E_X(s) \leq \delta_X^{-1} \frac{1}{|\Omega_X|} \sum_{s \in \Omega_X} E_X(s). \quad (161)$$

The factor δ_X^{-1} is best possible without additional information on the placement of $\mathcal{S}_X$.

Proof. Positivity gives $\sum_{s \in \mathcal{S}_X} E_X(s) \leq \sum_{s \in \Omega_X} E_X(s)$, which is (161) after division. Equality in the loss is attained by a function supported on $\mathcal{S}_X$. $\square$

For the family in (160), $\delta_X \asymp (\log X)^{-3}$ inside a fixed-proportion dyadic modulus window. Therefore a dense nonnegative estimate

$$\frac{1}{|\Omega_X|} \sum_{s \in \Omega_X} E_X(s) \ll \frac{\mathsf{T}_X}{(\log X)^A} \quad (162)$$

implies only

$$\frac{1}{|\mathfrak{M}_X|} \sum_{M \in \mathfrak{M}_X} E_X(M) \ll \frac{\mathsf{T}_X}{(\log X)^{A-3}}. \quad (163)$$

Thus $A > 3$ is the exact black-box threshold for a residual logarithmic saving, $A \geq 3 + \sigma$ is needed for a prescribed $(\log X)^{-\sigma}$ saving, and a qualitative $o(\mathsf{T}_X)$ conclusion on the sparse family requires a dense bound $o(\mathsf{T}_X/(\log X)^3)$. The same calculation applies to any nonnegative average of the canonical Hilbert defect. It is a selection threshold, not an estimate for that defect.

10.3 What dense shift estimates do and do not select

The averaged Chowla theorem of Matomäki et al. [9] averages over full boxes of integer shifts. Polynomial variants also exist: Frantzikinakis [3] treats independent polynomial shifts in a qualitative uniform-density framework. These results are important prior art for product-shaped shift parameters. A prime-restricted parameter box is a zero-density subset of the full integer box, so a qualitative dense convergence statement alone cannot select it. Stronger exceptional-set theorems must, however, be considered separately.

For shifted primes, the comparison can be made numerical. If $H = X^\vartheta$, $0 < \vartheta < 1$, then Lichtman [7, Theorem 1.1] proves, for every $\eta > 0$,

$$\sum_{h \leq H} \left| \sum_{p \leq X} \mu(p+h) \right| \ll_{\vartheta, \eta} \frac{H \pi(X)}{(\log X)^{1/3-\eta}}. \quad (164)$$

Restricting this nonnegative sum to a prescribed shift set of relative density $(\log X)^{-3}$ loses $(\log X)^3$ by lemma 10.1. The resulting average is bounded only by $\pi(X)(\log X)^{8/3+\eta}$, not by $o(\pi(X))$. In exponent language, the available dense saving is $1/3 - \eta$, while black-box selection of the triple-prime subset requires a saving strictly greater than 3: the deficit of this particular first-moment bound is $8/3 + \eta$. This calculation is not a limitation of Lichtman's full result.

Indeed, Lichtman [7, Theorem 6.3] proves a much sharper exceptional-set statement. Write $\log_2 X = \log \log X$ and suppose $\log H / \log_2 X \rightarrow \infty$. For fixed $h_2, \dots, h_k \leq H$, $g_1 = \mu$, and fixed 1-bounded $g_2, \dots, g_k$, the shifted-prime correlation is $o(\pi(X))$ for all but

$$O_K(H(\log X)^{-K}) \quad (165)$$

values of $h_1 \leq H$, for every $K > 0$. Consequently, taking $K > 3$ does select a good shift from any prescribed scalar shift set of size $\gg H/(\log X)^3$. Corollary 1.5 of the same paper gives the corresponding arbitrary logarithmic exceptional saving for fixed-order scalar Möbius tuples. Thus selection of a triple-prime subset in a pure scalar shifted-prime problem is already available and is not the TPC-42 gap.

The hybrid theorem of Lichtman and Teräväinen [8], in its stated parameter ranges, further inserts fixed Möbius shifts and fixed prime offsets and obtains cancellation for almost all choices of one shift once $H \geq (\log X)^{\ell+\varepsilon}$. What neither theorem supplies is one estimate uniform over the TPC family of coefficient fields: the modulus changes the residue fibers and their canonical

projector, and every nonzero alias changes the terminal argument. The row slopes, projective masks, and retained Hilbert coordinates are additional structure absent from the scalar theorem even when they are held fixed; they may also vary if a regular subpacket is reselected with the modulus. In particular, a scalar good shift chosen for one fixed collection of g_j does not bound the nonnegative moving quantity $\|T_{G_M}(I - \Pi_{w,M})\mu\|_{\mathcal{H}}^2$ averaged over M .

10.4 The claim boundary

The positive result of [section 6](#) is exactly a dimension-free postcomposition of the scalar residue vector already proved in TPC-41. Its useful new role in the present paper is to transfer, simultaneously, the inherited estimate through every residue-synchronous Bessel synthesis and to locate the within-fiber directional defect, whose physical synthesis appears as cross-residue coherence. It is not a vector extension of the arithmetic theorem of Klurman-Mangerel-Teräväinen, and it does not prove a new sparse modulus or sparse shift mean-value theorem. Conversely, [lemma 10.1](#) shows exactly what a future route must add: either direct information on the declared triple-prime family, or more than three powers of logarithmic saving in a nonnegative ambient average of that same canonical defect, before it can be selected by restriction alone. Scalar shift estimates, even with a smaller exceptional set, are not such a defect estimate.

11 The relocated gate

The two candidate shortcuts considered in TPC-41 now have exact endpoints. The scalar progression direction postcomposes through every bounded residue-synchronous Bessel map on the reduced-residue sector, but the actual terminal vectors are orthogonal rather than synchronous within each residue fiber. The localized convolution completes exactly, but its boundary is the negative local cell rather than a smaller error. Neither conclusion is heuristic.

11.1 Changing only the alias modulus does not move the equality column

Fix a common inherited physical row set, coefficient field, and terminal support, and vary only the auxiliary modulus used in the alias shifts. Attach that modulus to the notation and write $\mathcal{T}_k^L(M)$ for the output in (21). The coefficients c^L are inherited before the additive alias bank is introduced, and kM is the only occurrence of M in the terminal target.

Proposition 11.1 (Equality-column invariance). *For every two admissible auxiliary triples M, M' applied to this same physical coefficient field and support,*

$$\boxed{\mathcal{T}_0^L(M) = \mathcal{T}_0^L(M'), \quad \mathcal{T}_0^R(M) = \mathcal{T}_0^R(M').} \quad (166)$$

Consequently, for every nonempty finite admissible family $\mathfrak{M}$,

$$\frac{1}{|\mathfrak{M}|} \sum_{M \in \mathfrak{M}} (\|\mathcal{T}_0^L(M)\|^2 + \|\mathcal{T}_0^R(M)\|^2) = \|\mathcal{T}_0^L\|^2 + \|\mathcal{T}_0^R\|^2. \quad (167)$$

Proof. At $k = 0$, $n_{\alpha,j,0} = m_{\alpha}j + h$, independently of M . The coefficient in (20) is also independent of the auxiliary triple. This proves the two vector identities and hence their averaged norm identity. $\square$

Thus an average of a nonnegative complete physical energy over auxiliary triples can be useful only if it directly bounds the common equality column or supplies information on an M -dependent defect that can be transferred back to it. Merely selecting the smallest member of the average cannot make the equality column smaller. Nonzero aliases do depend on M , so a future

averaged theorem may still use them; the proposition closes only the naive equality-selection argument.

The common-field qualification is substantive. If a construction reselects residue masks, active rows, or a regular subpacket separately for each modulus, then those data are part of the M -dependence and [proposition 11.1](#) does not apply. The all-residue identities of [section 4](#) avoid such an implicit pruning.

11.2 The exact positive continuation

After projective separation, the direct route is the restricted orbit-slope Bessel estimate [\(112\)](#). In expanded form, its off-diagonal is a weighted average of the affine products

$$\mu(d_\alpha)\mu(d_\beta)\mu(m_\alpha j + h)\mu(m_\beta j + h), \quad \alpha \neq \beta, \quad (168)$$

with the actual TPC-36 masks and source weights. The two target affine forms are nonproportional, but their slopes and coefficients grow with X . Existing fixed-form or dense-shift theorems do not supply the uniform weighted estimate.

There are three now sharply separated subroutes.

- (a) **Direct cross-residue dispersion.** Prove [\(112\)](#) for the declared projective layers by a bilinear or Bessel inequality that uses the source signs, the affine determinants, and the orbit average jointly. This is the shortest route to the physical equality column.
- (b) **Average the canonical defect itself.** For each admissible triple, estimate

$$\|T_G(I - \Pi_w)\mu\|^2 \quad (169)$$

rather than a scalar residue variance. Positivity can then select a good triple for an M -dependent alias family. By [lemma 10.1](#), restriction from a dense modulus or shift average needs more than three powers of logarithmic saving unless the theorem is proved directly on the triple-prime family.

- (c) **Recombine localization before taking the Hilbert norm.** The three-face completion of [theorem 8.1](#) cannot treat one local cell. A different argument could seek cancellation among a complete signed partition of all dyadic cells *before* squaring. Such a theorem must retain the physical (γ, j) coordinates; scalar convolution after those coordinates have been summed is insufficient.

The first route is the next minimal theorem. The second is a genuine selection problem with an explicit density threshold. The third changes the order of localization and Hilbert synthesis and is therefore a different architecture, not a repair of the small-boundary argument.

11.3 Claim boundary

This paper proves no bound of the form [\(112\)](#) or [\(113\)](#). It does not prove cancellation for an individual pair of growing affine forms, a fixed-shift Chowla theorem, a diagonal-scale raw scalar progression variance, or the physical four-Möbius estimate [\(30\)](#). The countermodels rule out universal black-box implications; they do not assert that the actual Möbius coefficients saturate those models. The convolution lower bound concerns an unmasked coefficient cell; it is not a lower bound for the signed terminal output. In particular, no sieve-parity breach, Hardy-Littlewood asymptotic, or infinitude of twin primes is claimed.

12 Exact certificate and claim ledger

The arguments above are analytic or exact finite-dimensional identities. The companion standard-library certificate checks the finite identities with integers, Gaussian integers, rational arithmetic, and exact finite Dirichlet convolutions. It uses no floating point, random draws, or optimization-sensitive assertion.

12.1 Certified identities

Run from the paper directory

```
python experiments/tpc42_certificate.py
python -O experiments/tpc42_certificate.py
```

The two commands produce byte-identical canonical JSON. The frozen run passes

$$\boxed{178,226 \text{ exact checks.}} \quad (170)$$

It verifies:

- (i) injectivity of the row-to-residue map on every declared finite orbit slice;
- (ii) fixed-alias orthogonality for distinct products in one residue fiber, including nonunit fibers, the exact packed diagonal, and the cross-residue coherent decomposition;
- (iii) the corresponding minimal-folded orthogonality, packed energy, coherent energy, and ordered cross-residue identity;
- (iv) the $1/\rho$ synchronous energy and $1 - 1/\rho$ canonical defect of orthogonal fibers, together with the sharp zero-scalar countermodels;
- (v) the three-face convolution identity, the physical-window counterterm, outer-prime uniqueness, prime-power support separation, and exact coefficient-energy factorization through the frozen range; and
- (vi) every endpoint rational-exponent identity, including $Q/K_B = X^{o(1)}J^2$.

The certificate digest is

c2860f5d7c679224a5ff4edd417cb956a3cb49e3c7addc434e23dd7b4b8c50bf

The SHA-256 hashes of the canonical JSON record and source are, respectively,

1f6d7defff20b4ee8dfd4cd75b86c39eb08492a7a021e6e569dabc5aeebaf51a
57e1438392c88e8523bd5e038f580f179cdcd7f0590776507e2956202c9c52

12.2 What the certificate does not certify

The finite checks do not verify the prime number theorem, the multiplicative large sieve, the TPC-41 scalar progression theorem, or any asymptotic inherited from the previous papers. They do not test Möbius randomness and are not numerical evidence for cancellation in (168). The hard-cell convolution model checks an exact coefficient identity; it does not model every physical mask.

In particular, the JSON record marks as false any interpretation that it proves the orbit-slope Bessel target, a fixed affine Chowla estimate, a parity breach, a prime-pair asymptotic, or the twin-prime conjecture. Its purpose is reproducibility of the exact structural reductions and their claim boundary.

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